M. Rohaan 221-2327 Sec-E A#3

Q#01:-

a) Given Networks

185.34.52.0/26
185.34.52.64/26
185.34.52.128/25
185.34.53.0/24

Step #01:-

185.34.52.0/26

* prefix = /26

Blocksize = $2^{32-prefix} = 2^{32-26} = 2^6 = 64$ IPs

Starting IP = 185.34.52.0

Range: 185.34.52.0 – 185.34.52.0 + 63
→ 185.34.52.0 – 185.34.52.63

Step #02:-

185.34.52.64/26

* prefix = /26

Blocksize = $2^{32-26} = 2^6 = 64$ IPs

Starting IP = 185.34.52.64

Range: 185.34.52.64 – 185.34.52.64 + 63
→ 185.34.52.64 – 185.34.52.127

Step #03:-

185.34.52.128/25

* prefix = /25 → block size = $2^{32-25} = 2^7 = 128$ IPs

Starting IP = 185.34.52.128

Range: 185.34.52.128 — 185.34.52.128+127

→ 185.34.52.128 — 185.34.52.255

All these ranges have no gaps, together

→ 185.34.52.0 — 185.34.52.255

∴ exactly one full network /24, because /24 has block size $2^{32-24} = 2^8 = 256$ addresses from 0 - 255.

Step #04:-

185.34.53.0/24

prefix = /24 → block size = $2^{32-24} = 2^8 = 256$

Starting IP = 185.34.53.0

Range: 185.34.53.0 — 185.34.53.255

also /24 network 256 IPs (0 — 255)

Step #05:-

Combine the two /24s

185.34.52.0/24 → 185.34.52.0 — 185.34.52.255

185.34.53.0/24 → 185.34.53.0 — 185.34.53.255

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Together they cover,

185.34.52.0 — 185.34.53.255

∴ Two consecutive /24 blocks combine into /23 because /24 = 256 address, Two of them = 512 addressess, So /23, blocksize = $2^{32-23} = 2^9 = 512$ addressess.

Smallest single aggregate block is 185.34.52.0/23

b) Given

198.51.120.128/26

Step #01:-

prefix = /26 → blocksize = $2^{32-26} = 2^6 = 64$ IPs

Range : 198.51.120.128 — 198.51.120.128

→ 198.51.120.128 — 198.51.120.191

Network address : 198.51.120.128

Broadcast address : 198.51.120.191

Host Range : 198.51.120.129 — 198.51.120.190

198.51.120.130 is a valid host IP.

c) Given

198.51.120.0/24

Step #01:-

Prefix = /24 → Blocksize = $2^{32-24} = 2^8 = 256$ IPs

* Class C

Subnets Needed (Upper Bound)

$2^3 \geq 5$

Subnet of the class of IP address

255.255.255.0

11111111.11111111.11111111.00000000

Step #02:-

Total required: $2^3 = 8$

Class C capacity: $2^8 = 256$

$8 \ll 256$

Subnetting possible

Step #03:-

Upper Bound 2^3 (Borrow 3 bits from Network)

11111111.11111111.11111111.11100000

255.255.255.224

Step #04:-

256 - 224 = 32

Subnetting by each block size of 32.

Step #05:-

1) 198.51.120.0/27 (0-31)
2) 198.51.120.32/27 (32-63)
3) 198.51.120.64/27 (64-95)
4) 198.51.120.96/27 (96-127)
5) 198.51.120.128/27 (128-159)
6) 198.51.120.160/27 (160-191)
7) 198.51.120.192/27 (192-223)
8) 198.51.120.224/27 (224-255)

Five equal sized subnets are, while remaining 3 will be unused or used by later.

Q#02:-

a)

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